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Subject Name: Report Project 1 - Probability and Statistics

23 June 2026

1. Problem statement

This project focuses on a dataset of 1,000 emails and aims to evaluate the probability that an email is spam or legitimate. The analysis investigates whether the presence of the word “FREE” can serve as an indicator of spam by examining the relationship between this feature and the email classification.

2. Objectives

From the problem, the dataset includes:

- Total: 1000 emails
- Spam: 300 emails
- Legitimate: 700 emails
- Spam with “FREE”: 240 emails
- Legitimate with “FREE”: 70 emails

From the dataset above, we would like to compute basic analysis statistics: marginal, joint and condition probability. From those foundations, we are going to point out the possibility of an event with the given condition. Moreover, we are evaluating the relationship between spam email and the word “FREE”. Then, I will propose some statistical numbers including Accuracy, Precision, ... through the confusion matrix and the heatmap created by Python scripts.

3. Computation and Analysis

Before going to the actual computations, we arrange the data into one table:

	F (For FREE)	~F (For not FREE)	Total
S (For Spam)	240	60	300
L (For Legitimate)	70	630	700
Total	310	690	1000

From then, we can compute the marginal as:

$$P(S) = \frac{300}{1000} = 0.3$$

$$P(L) = \frac{700}{1000} = 0.7$$

$$P(F) = \frac{310}{1000} = 0.31$$

$$P(\sim F) = \frac{690}{1000} = 0.69$$

Similarly, we can compute for joint probabilities:

$$P(S \cap F) = \frac{240}{1000} = 0.24$$

$$P(S \cap \sim F) = \frac{60}{1000} = 0.06$$

$$P(L \cap F) = \frac{70}{1000} = 0.07$$

$$P(L \cap \sim F) = \frac{630}{1000} = 0.63$$

As well as for conditional probabilities:

$$P(S|F) = \frac{240}{310} = 0.7741935$$

$$P(S|\sim F) = \frac{60}{690} = 0.0869565$$

$$P(L|F) = \frac{70}{310}$$

$$P(L|\sim F) = \frac{630}{690}$$

Also:

$$P(F|S) = \frac{240}{300}$$

$$P(\sim F|S) = \frac{60}{300}$$

$$P(F|L) = \frac{70}{700}$$

$$P(\sim F|L) = \frac{630}{700}$$

Based on $P(S|F) = 0.7742 = 77.42\%$ It means 77.42% of emails that contain “FREE” are spam emails.

- Addition Rule

The addition rule is used to find the probability that an email is either spam, contains the word FREE, or both.

$$P(S \cup F) = P(S) + P(F) - P(S \cap F)$$

Where:

- $P(S) = 0.3$
- $P(F) = 0.31$
- $P(S \cap F) = 0.24$

Substituting these values:

$$P(S \cup F) = 0.3 + 0.31 - 0.24 = 0.37$$

Interpretation

The probability that an email is either spam or contains the word FREE (or both) is 0.37, or 37%.
This means that 370 out of the 1,000 emails belong to at least one of these categories.

- Multiplication Rule

The multiplication rule is used to determine the probability that an email is both spam and contains the word FREE.

$$P(S \cap F) = P(F|S) * P(S)$$

From the dataset:

$$P(F|S) = 0.80$$

$$P(S) = 0.30$$

Substituting:

$$P(S \cap F) = 0.80 \times 0.30 = 0.24$$

The result matches the joint probability obtained directly from the dataset:

Interpretation

The probability that an email is both spam and contains the word FREE is 0.24, or 24%. In the dataset, 240 out of the 1,000 emails satisfy both conditions.

4. Independence test

Connection to Independence Test

To determine whether Spam and FREE are independent events:

$$P(S) * P(F) = 0.093$$

However,

$$P(S \cap F) = 0.24$$

Since

$$0.24 \neq 0.0930$$

The events "is spam" and "contains FREE" are not independent. The occurrence of the word FREE significantly affects the probability that an email is spam. This suggests that FREE is a useful feature for spam classification.

5. False Positive and False Negative

False Positive here I defined as the email containing "FREE" but are not the spam emails.

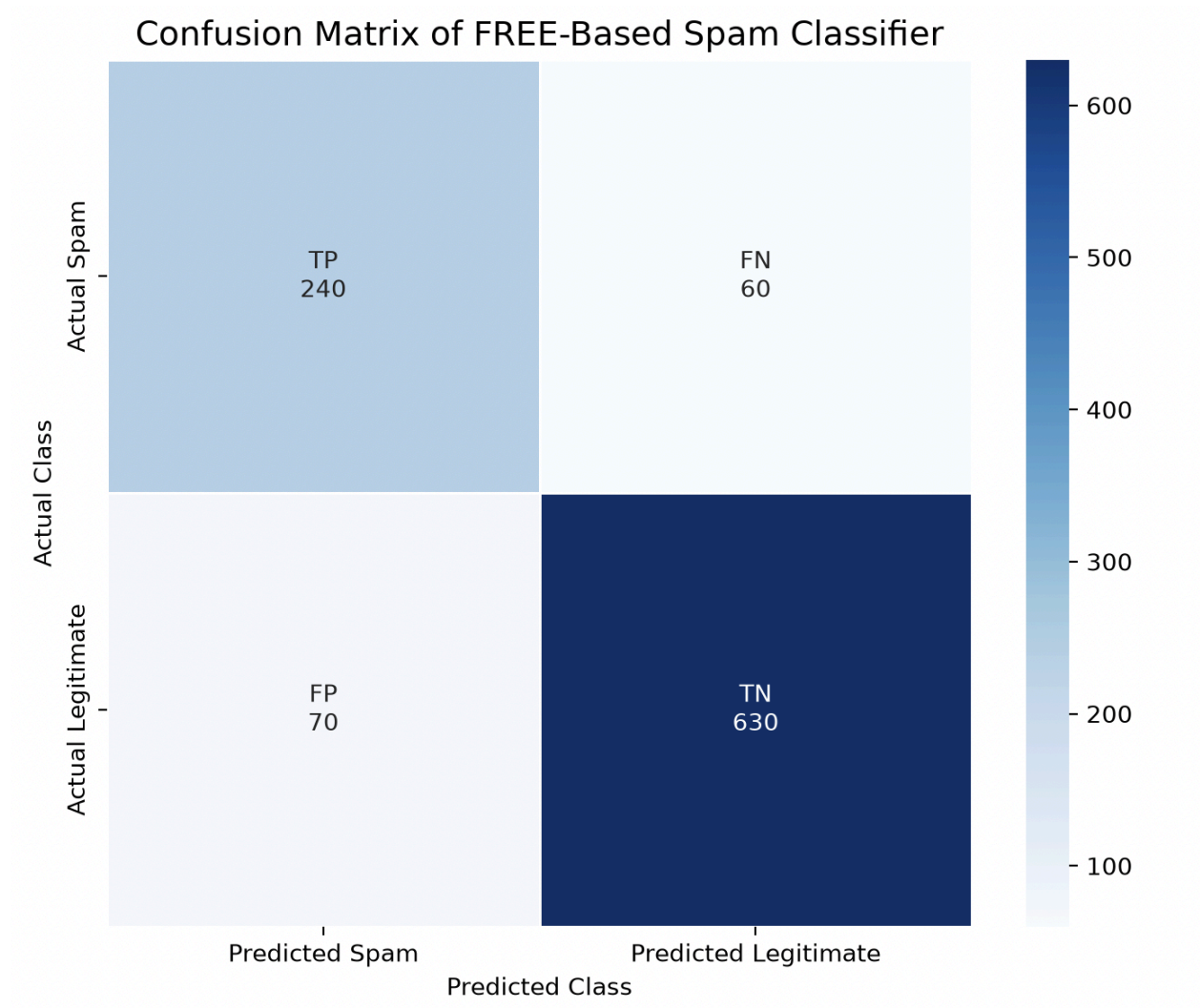
False Negative is the email that does not contain "FREE" but are the spam emails.

So, based on that, we can see from the table:

False Positive: 70

False Negative: 60

6. Performance matrix



$$Accuracy = \frac{TP+TN}{1000} = \frac{240+630}{1000} = 0.87$$

$$Precision = \frac{TP}{TP+FP} = \frac{240}{240+70} = 0.7742$$

$$Recall = \frac{240}{240+60} = 0.80$$

